

NAME: NISHI CHOPDA
JEFF BEZOS
150096725128

Excel Sheet

https://docs.google.com/spreadsheets/d/1FMCmQDhs-E6NY3ygimtVmCXfAB4iV_FSLI0zgqZFm3A/edit?usp=sharing

Westfold Textile Mills – Multi-Site Routing, Access Control & Address Translation

1. Introduction

Westfold Textile Mills consists of a Headquarters (HQ), a Weaving Plant, and the recently acquired Loomcraft Fabrics network. The network is designed to provide communication between these sites while maintaining the existing EIGRP routing environment at Loomcraft Fabrics.

The proposed network uses OSPF for internal routing between the HQ and Plant, EIGRP for the Loomcraft network, and BGP for redundant connections to two Internet Service Providers. Network security is implemented using VLANs and Access Control Lists (ACLs) to restrict communication between Finance and Guest networks. NAT/PAT is used to provide Internet access for internal devices.

2. Objectives

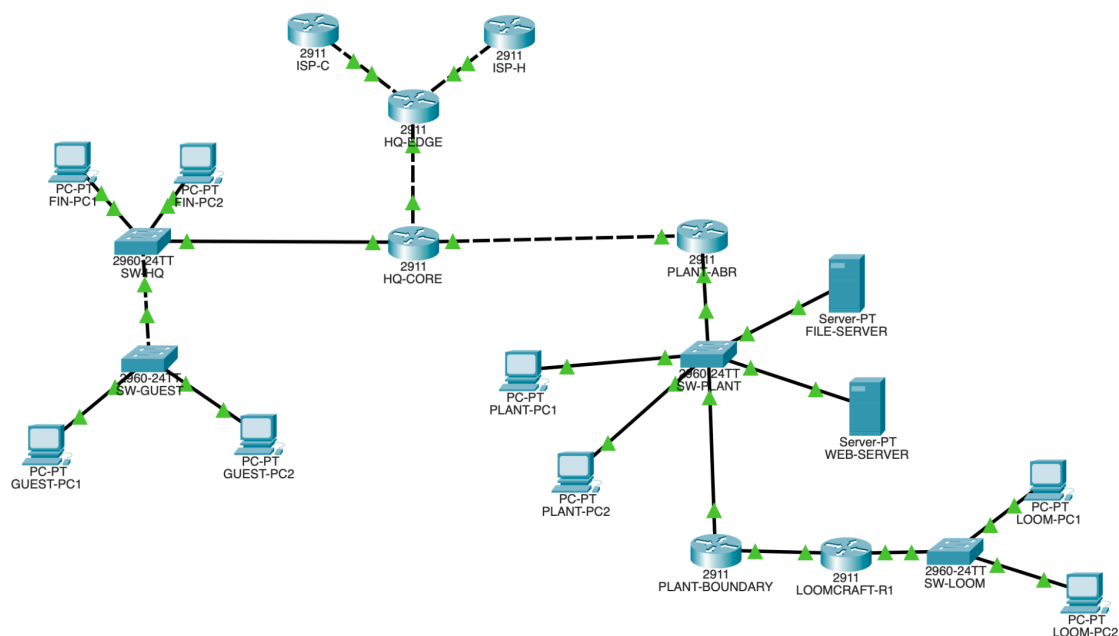
The main objectives of this project are:

1. Configure static and default routing as the initial routing configuration.
2. Replace the HQ–Plant static routes with multi-area OSPF.
3. Configure OSPF Area 0 and Area 10 and verify neighbor adjacency and DR/BDR election.

4. Maintain the existing Loomcraft EIGRP network using EIGRP AS 950.
5. Configure route redistribution and route summarization between OSPF and EIGRP.
6. Configure eBGP connections with ISP-Crestline and ISP-Harbor.
7. Configure ACLs to restrict Guest access to the Finance network while allowing required communication.
8. Configure NAT/PAT for internal Internet access.
9. Configure static NAT for the external access of an internal web service.
10. Verify routing, security, NAT, and failover behavior using appropriate Cisco IOS commands and connectivity tests.

3. Network Design

3.1 Network Topology



Network Topology – Westfold Textile Mills

The topology represents the network architecture of Westfold Textile Mills, including the Headquarters, Weaving Plant, Loomcraft Fabrics network, dual ISP connections, Finance and Guest VLANs, and Plant-based file and web servers.

3.2 Device List

The network topology consists of routers, switches and end devices distributed across the Westfold Textile Mills Headquarters, Plant and Loomcraft Fabrics network.

Device	Quantity	Purpose
Cisco 2911 Router	7	Routing, OSPF, EIGRP, BGP, NAT/PAT and Internet connectivity
Cisco 2960-24TT Switch	4	LAN connectivity and VLAN implementation
Finance PCs	2	Finance department systems
Guest PCs	2	Guest network users
Plant PCs	2	Plant network users
Loomcraft PCs	2	Loomcraft Fabrics users
File Server	1	Shared file services
Web Server	1	Internal web service for static NAT testing

Router roles:

HQ-EDGE provides Internet connectivity, BGP and NAT/PAT.
HQ-CORE provides internal HQ routing and Finance/Guest

VLAN gateways. PLANT-ABR connects the HQ backbone to the Plant network and provides the OSPF Area 0 to Area 10 boundary. PLANT-BOUNDARY connects the Plant OSPF network to the Loomcraft EIGRP network. LOOMCRAFT-R1 maintains the existing EIGRP AS 950 network. ISP-C and ISP-H represent the two Internet Service Providers.

3.3 IP Addressing

The network uses the private IPv4 address space **172.31.0.0/16** for internal communication. The address space is divided into smaller subnets according to the requirements of each network segment. /30 subnets are used for point-to-point router links, while /24 subnets are used for LANs and VLANs. Separate networks are maintained for Finance, Guest users, Plant and Loomcraft. The simulated ISP connections use separate transit networks, while the public block **198.51.100.24/29** is reserved for NAT/PAT.

Router Interface Addressing

Device	Interface	IP Address	Subnet Mask	Purpose
HQ-EDGE	G0/0	203.0.113.1	255.255.255.252	ISP-Crestline
HQ-EDGE	G0/1	203.0.113.5	255.255.255.252	ISP-Harbor
HQ-EDGE	G0/2	172.31.0.1	255.255.255.252	HQ-CORE

HQ-CORE	G0/0	172.31.0.2	255.255.255.252	HQ-EDGE
HQ-CORE	G0/1	172.31.0.5	255.255.255.252	PLANT-ABR
HQ-CORE	G0/2.15	172.31.15.1	255.255.255.0	Finance Gateway
HQ-CORE	G0/2.85	172.31.85.1	255.255.255.0	Guest Gateway
PLANT-ABR	G0/0	172.31.0.6	255.255.255.252	HQ-CORE
PLANT-ABR	G0/1	172.31.10.1	255.255.255.0	Plant LAN
PLANT-BOUNDARY	G0/0	172.31.20.1	255.255.255.252	LOOMCRAFT-R1
PLANT-BOUNDARY	G0/1	172.31.10.2	255.255.255.0	Plant LAN
LOOMCRAFT-R1	G0/0	172.31.20.2	255.255.255.252	PLANT-BOUNDARY

LOOMCRAFT-R1	G0/1	172.31.50.1	255.255.255.0	Loomcraft LAN
ISP-C	G0/0	203.0.113.2	255.255.255.252	HQ-EDGE
ISP-H	G0/0	203.0.113.6	255.255.255.252	HQ-EDGE

End Device Addressing

Device	IP Address	Default Gateway
FIN-PC1	172.31.15.10	172.31.15.1
FIN-PC2	172.31.15.11	172.31.15.1
GUEST-PC1	172.31.85.10	172.31.85.1
GUEST-PC2	172.31.85.11	172.31.85.1
PLANT-PC1	172.31.10.10	172.31.10.1
PLANT-PC2	172.31.10.11	172.31.10.1

FILE-SERVER	172.31.10.20	172.31.10.1
WEB-SERVER	172.31.10.21	172.31.10.1
LOOM-PC1	172.31.50.10	172.31.50.1
LOOM-PC2	172.31.50.11	172.31.50.1

Addressing Summary:

The internal addressing scheme provides separate networks for each major functional area while allowing routing between sites. Finance and Guest users are placed in separate VLANs to support later ACL-based access control. The Plant LAN provides a shared multi-access network for OSPF Area 10, while the Loomcraft network remains separate under EIGRP AS 950.

IP Addressing

Routers:

1. HQ-EDGE

Device	Interface	IP Address	Subnet Mask	Purpose
HQ-EDGE	G0/0	203.0.11 3.1	255.255.255. 252	Connection to ISP-Crestline
HQ-EDGE	G0/1	203.0.11 3.5	255.255.255. 252	Connection to ISP-Harbor

HQ-EDGE	G0/2	172.31.0. 1	255.255.255. 252	Connection to HQ-CORE
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HQ-EDGE Interface Configuration

Purpose:

HQ-EDGE acts as the Internet-edge router for Westfold Textile Mills. It provides connectivity to both ISP routers and connects the external network to the internal HQ network.

Configuration:

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 203.0.113.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 203.0.113.5 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#ip address 172.31.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Verification:

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  203.0.113.1    YES manual up          down
GigabitEthernet0/1  203.0.113.5    YES manual up          down
GigabitEthernet0/2  172.31.0.1     YES manual up          down
Vlan1           unassigned     YES unset  administratively down down
Router#
```

2. HQ-CORE

Device	Interface	IP Address	Subnet Mask	Purpose
HQ-CORE	G0/0	172.31.0.2	255.255.255.252	Connection to HQ-EDGE
HQ-CORE	G0/1	172.31.0.5	255.255.255.252	Connection to PLANT-ABR

HQ-CORE Interface Configuration

Purpose:

HQ-CORE acts as the central routing device for the HQ network. It connects the HQ Internet edge to the Plant backbone and will later provide gateway services for the Finance and Guest VLANs.

Configuration:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 172.31.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.31.0.5 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification:

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 172.31.0.2      YES manual up          up
GigabitEthernet0/1 172.31.0.5      YES manual up          down
GigabitEthernet0/2 unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
Router#
```

3. PLANT-ABR

Device	Interface	IP Address	Subnet Mask	Purpose
PLANT-ABR	G0/0	172.31.0.6	255.255.255.252	Connection to HQ-CORE

PLANT-ABR	G0/1	172.31.1 0.1	255.255.255. 0	Plant LAN / OSPF Area 10
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PLANT-ABR Interface Configuration

Purpose:

PLANT-ABR connects the HQ backbone to the Plant network. It will later function as the **OSPF Area Border Router (ABR)** between Area 0 and Area 10.

Configuration:

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 172.31.0.6 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.31.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Verification:

```

Router#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0       172.31.0.6      YES manual up              up
GigabitEthernet0/1       172.31.10.1     YES manual up              up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Router#

```

4. PLANT-BOUNDARY

Device	Interface	IP Address	Subnet Mask	Purpose
PLANT-BOUNDARY	G0/0	172.31.20.1	255.255.255.252	Connection to LOOMCRAFT-R1
PLANT-BOUNDARY	G0/1	172.31.10.2	255.255.255.0	Plant LAN / OSPF Area 10

PLANT-BOUNDARY Interface Configuration

Purpose:

PLANT-BOUNDARY connects the Plant LAN with the Loomcraft network. It will later act as the routing boundary between **OSPF** and **EIGRP AS 950** and will be used for route redistribution.

Configuration:

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 172.31.20.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.31.10.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Verification:

```

Router#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 172.31.20.1     YES manual up          down
GigabitEthernet0/1 172.31.10.2     YES manual up          up
GigabitEthernet0/2 unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
Router#

```

5. LOOMCRAFT-R1

Device	Interface	IP Address	Subnet Mask	Purpose
LOOMCRAFT-R1	G0/0	172.31.20.1 0.2	255.255.255.252	Connection to PLANT-BOUNDARY
LOOMCRAFT-R1	G0/1	172.31.10.2 0.1	255.255.255.0	Loomcraft LAN

LOOMCRAFT-R1 Interface Configuration

Purpose:

LOOMCRAFT-R1 provides connectivity between the Loomcraft network and the Plant boundary router. The router will later operate using **EIGRP AS 950**, preserving the existing Loomcraft routing domain.

Configuration:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 172.31.20.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.31.50.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification:

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       172.31.20.2     YES manual  up          up
GigabitEthernet0/1       172.31.50.1     YES manual  up          up
GigabitEthernet0/2       unassigned      YES unset   administratively down down
Vlan1                    unassigned      YES unset   administratively down down
Router#
```

PLANTS END DEVICES (PCS):

1. PLANT-PC1

Device	IP Address	Subnet Mask	Default Gateway	Purpose
PLANT-PC1	172.31.10.1 0	255.255.255. 0	172.31.10.1	Plant workstation

PLANT-PC1 Configuration

The screenshot displays the configuration interface for PLANT-PC1, organized into tabs: Physical, Config, Desktop (selected), Programming, and Attributes.

IP Configuration

- Interface: FastEthernet0
- IP Configuration:
 - ☐ DHCP
 - ☒ Static
 - IPv4 Address: 172.31.10.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 172.31.10.1
 - DNS Server: 0.0.0.0

IPv6 Configuration

- ☐ Automatic
- ☒ Static
- IPv6 Address: [Empty field]
- Link Local Address: FE80::20A:41FF:FE9A:7781
- Default Gateway: [Empty field]
- DNS Server: [Empty field]

802.1X

- ☐ Use 802.1X Security
- Authentication: MD5
- Username: [Empty field]
- Password: [Empty field]

At the bottom left, there is a "Top" link.

Verification:

The PC successfully pinged its default gateway with **0% packet loss**, confirming local network connectivity.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.10.1

Pinging 172.31.10.1 with 32 bytes of data:

Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

2. PLANT-PC2

Device	IP Address	Subnet Mask	Default Gateway	Purpose
PLANT-PC2	172.31.10.11	255.255.255.0	172.31.10.1	Plant workstation

PLANT-PC2 Configuration

The screenshot shows a network configuration window titled "IP Configuration" with a tabbed interface. The "Desktop" tab is selected. The "Interface" dropdown is set to "FastEthernet0".

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 172.31.10.11

Subnet Mask: 255.255.255.0

Default Gateway: 172.31.10.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: [empty] / [empty]

Link Local Address: FE80::201:97FF:FEA3:59EB

Default Gateway: [empty]

DNS Server: [empty]

802.1X

☐ Use 802.1X Security

Authentication: MDS

Username: [empty]

Password: [empty]

☐ Top

Verification:

PLANT-PC2 successfully reached its default gateway with **0% packet loss**.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.10.1

Pinging 172.31.10.1 with 32 bytes of data:

Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```



Server:

1. FILE-SERVER

Device Addressing

Device	IP Address	Subnet Mask	Default Gateway	Purpose
FILE-SERVER	172.31.10.20	255.255.255.0	172.31.10.1	Shared file services

FILE-SERVER Configuration

Purpose:

FILE-SERVER is connected to the Plant LAN and provides shared file services for the network. It is configured with a static IP address so that it can be consistently accessed by other devices.

IP Configuration

Physical Config Services **Desktop** Programming Attributes

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.31.10.20

Subnet Mask 255.255.255.0

Default Gateway 172.31.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20A:F3FF:FE89:EC09

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

Verification

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 172.31.10.1

Pinging 172.31.10.1 with 32 bytes of data:

Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

2. WEB-SERVER

Device	IP Address	Subnet Mask	Default Gateway	Purpose
WEB-SERVER	172.31.10.21	255.255.255.0	172.31.10.1	Internal web service used for Static NAT testing

Configuration:

The WEB-SERVER was assigned a static IP address in the Plant LAN. The default gateway is the PLANT-ABR router interface (172.31.10.1). The HTTP service was enabled so that the server can later be accessed externally through Static NAT.

The screenshot displays the 'IP Configuration' window in a network management interface. The window has tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'Desktop' tab is selected. The 'IP Configuration' section is active, showing the following settings:

- IP Configuration:**
 - ☐ DHCP
 - ☒ Static
 - IP Address: 172.31.10.21
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 172.31.10.1
 - DNS Server: 0.0.0.0
- IPv6 Configuration:**
 - ☐ Automatic
 - ☒ Static
 - IPv6 Address: [Empty field]
 - Link Local Address: FE80::2E0:8FFF:FE8A:9E9E
 - Default Gateway: [Empty field]
 - DNS Server: [Empty field]
- 802.1X:**
 - ☐ Use 802.1X Security
 - Authentication: MD5
 - Username: [Empty field]
 - Password: [Empty field]

At the bottom left, there is a 'Top' button.

Verification:

A ping test from the WEB-SERVER to 172.31.10.1 was successful

with **0% packet loss**, confirming local connectivity to the Plant gateway.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 172.31.10.1

Pinging 172.31.10.1 with 32 bytes of data:

Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

LOOMCRAFT END DEVICES (PCS)

LOOM-PC1

Device	IP Address	Subnet Mask	Default Gateway	Purpose
LOOM-PC1	172.31.50.10	255.255.255.0	172.31.50.1	Loomcraft workstation

LOOM-PC1 Configuration

LOOM-PC1 was configured with a static IP address in the Loomcraft LAN. The default gateway was set to 172.31.50.1, which is the LAN interface of LOOMCRAFT-R1.

The screenshot shows the 'IP Configuration' window in Cisco Packet Tracer. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IPv4 Address: 172.31.50.10, Subnet Mask: 255.255.255.0, Default Gateway: 172.31.50.1, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The IPv6 Address field is empty, while the Link Local Address is set to FE80::201:C9FF:FEE4:8277. The '802.1X' section has 'Use 802.1X Security' unchecked, and the 'Authentication' dropdown is set to 'MD5'. The 'Username' and 'Password' fields are empty. A 'Top' button is located at the bottom left of the window.

Verification:

The PC successfully pinged its default gateway 172.31.50.1 with **0% packet loss**, confirming local connectivity.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.50.1

Pinging 172.31.50.1 with 32 bytes of data:

Reply from 172.31.50.1: bytes=32 time=1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.50.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

2. LOOM-PC2

Device	IP Address	Subnet Mask	Default Gateway	Purpose
LOOM-PC2	172.31.50.11	255.255.255.0	172.31.50.1	Loomcraft workstation

LOOM-PC2 Configuration

Configuration:

LOOM-PC2 was configured with a static IP address of 172.31.50.11, subnet mask 255.255.255.0, and default gateway 172.31.50.1. The device belongs to the Loomcraft Fabrics LAN.

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

Static

IPv4 Address172.31.50.11

Subnet Mask255.255.255.0

Default Gateway0.0.0.0

DNS Server172.31.50.1

IPv6 Configuration

Static

IPv6 Address

Link Local AddressFE80::2D0:BAFF:FE55:4692

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMDS

Username

Password

Top

Verification:

LOOM-PC2 successfully reached its default gateway 172.31.50.1 with **0% packet loss**, confirming successful local network connectivity.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.50.1

Pinging 172.31.50.1 with 32 bytes of data:

Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.50.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

ISP-C

Device	Interface	IP Address	Subnet Mask	Purpose
ISP-C	G0/0	203.0.113.2	255.255.255.252	Connection to HQ-EDGE

ISP-C Interface Configuration

Purpose:

ISP-C represents the **Crestline Internet Service Provider** in the network topology. Its GigabitEthernet0/0 interface is configured with

the IP address 203.0.113.2/30, which connects it to the HQ-EDGE router through the ISP-C transit network.

Configuration:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 203.0.113.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification:

The ISP-C interface was verified using show ip interface brief. The GigabitEthernet0/0 interface shows the configured IP address 203.0.113.2 and an **up/up** status, confirming that the interface is operational.

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  203.0.113.2    YES manual up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
Router#
```

ISP-H

Device	Interface	IP Address	Subnet Mask	Purpose
ISP-H	G0/0	203.0.113.6	255.255.255.252	Connection to HQ-EDGE

ISP-H Interface Configuration

Purpose:

ISP-H represents the **Harbor Internet Service Provider** in the network topology. Its GigabitEthernet0/0 interface is configured with the IP address 203.0.113.6/30, which connects it to the HQ-EDGE router through the ISP-H transit network.

Configuration:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 203.0.113.6 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification:

The ISP-H interface was verified using show ip interface brief. The GigabitEthernet0/0 interface shows the configured IP address 203.0.113.6 and an **up/up** status, confirming that the interface is operational.

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       203.0.113.6     YES manual up          up
GigabitEthernet0/1       unassigned      YES unset   administratively down down
GigabitEthernet0/2       unassigned      YES unset   administratively down down
Vlan1                    unassigned      YES unset   administratively down down
Router#
```

SW-HQ VLAN Configuration

Purpose:

SW-HQ is used to provide Layer 2 connectivity for the Finance and Guest networks. Two separate VLANs are created to logically isolate the Finance and Guest users.

VLAN ID	VLAN Name	Network	Purpose
15	FINANCE	172.31.15.0/24	Finance users
85	GUEST	172.31.85.0/24	Guest users

Configuration:

```
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 15
Switch(config-vlan)#name FINANCE
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 85
Switch(config-vlan)#name GUEST
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification:

The show vlan brief command confirms that VLAN 15 (FINANCE) and VLAN 85 (GUEST) have been successfully created and are in the **active** state.

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
15	FINANCE	active	
85	GUEST	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Finance Port Configuration

The two Finance PCs are connected to SW-HQ through FastEthernet ports Fa0/2 and Fa0/3. These ports are configured as access ports and assigned to VLAN 15 (FINANCE). This ensures that both Finance workstations belong to the Finance network.

Configuration

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface range fastEthernet 0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 15
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification

The show vlan brief command was used to verify the port assignments. The output confirms that **Fa0/2 and Fa0/3 are assigned to VLAN 15 (FINANCE)** and the VLAN is active.

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2 Fa0/2, Fa0/3
15	FINANCE	active	
85	GUEST	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Trunk Configuration

Theory:

A trunk port is used to carry traffic from multiple VLANs over a single physical link. In this network, the connection between SW-HQ and HQ-CORE must carry both the Finance VLAN (VLAN 15) and Guest VLAN (VLAN 85). The trunk uses IEEE 802.1Q VLAN tagging so that the router can identify the VLAN associated with each frame.

Purpose:

Fa0/1 connects SW-HQ to the HQ-CORE router. The port is configured as a trunk so that VLAN 15 and VLAN 85 traffic can reach the corresponding router subinterfaces:

- VLAN 15 → G0/2.15 → 172.31.15.1
- VLAN 85 → G0/2.85 → 172.31.85.1

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 15,85
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification:

The show interfaces trunk command confirms that Fa0/1 is operating in **trunking** mode using **802.1Q** encapsulation. VLANs **15 and 85** are allowed on the trunk.

```
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking     1

Port      Vlans allowed on trunk
Fa0/1     15,85

Port      Vlans allowed and active in management domain
Fa0/1     15,85

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     85

Switch#
```

SW-HQ to SW-GUEST Trunk Configuration

Theory:

A trunk link is used to carry VLAN traffic between switches while maintaining VLAN identification through IEEE 802.1Q tagging. In this network, the link between SW-HQ and SW-GUEST carries Guest VLAN 85 so that Guest devices can communicate with their gateway through HQ-CORE.

Configuration on SW-HQ:

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 85
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Configuration on SW-GUEST:

```

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 85
Switch(config-vlan)#name GUEST
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk allowed vlan 85
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

```

Verification:

The show interfaces trunk command confirms that SW-HQ Fa0/4 is operating in **trunking mode** using **802.1Q encapsulation**. VLAN 85 is allowed on the trunk and is in the spanning-tree forwarding state. The corresponding SW-GUEST Fa0/1 interface is also configured as a trunk carrying VLAN 85.

```

Switch#show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1
Fa0/4	on	802.1q	trunking	1


```

Port          Vlans allowed on trunk
Fa0/1         15,85
Fa0/4         85

Port          Vlans allowed and active in management domain
Fa0/1         15,85
Fa0/4         85

Port          Vlans in spanning tree forwarding state and not pruned
Fa0/1         15,85
Fa0/4         85

Switch#

```

Guest Port Configuration

The Guest PCs are connected to SW-GUEST through FastEthernet ports Fa0/2 and Fa0/3. These ports are configured as access ports and assigned to VLAN 85 (GUEST). This ensures that both Guest workstations belong to the dedicated Guest network.

Configuration

```
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface range fastEthernet 0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 85
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification

The show vlan brief command was used to verify the Guest port assignments. The output confirms that **Fa0/2 and Fa0/3 are assigned to VLAN 85 (GUEST)** and the VLAN is active.

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
85	GUEST	active	Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Finance PCs

The Finance PCs are assigned to the **Finance VLAN (VLAN 15)**. The Finance network uses the 172.31.15.0/24 subnet, with the HQ-CORE router providing the default gateway through 172.31.15.1.

Device	IP Address	Subnet Mask	Default Gateway	Purpose
FIN-PC1	172.31.15.10	255.255.255.0	172.31.15.1	Finance workstation
FIN-PC2	172.31.15.11	255.255.255.0	172.31.15.1	Finance workstation

Finance PC Configuration

The Finance PCs are configured with static IPv4 addresses belonging to the Finance VLAN. The default gateway is set to 172.31.15.1, which allows the PCs to communicate with other permitted networks through HQ-CORE.

Purpose:

The Finance PCs represent the Finance department systems. Their network will later be protected using ACLs so that Guest users cannot initiate connections to Finance resources.

Physical

Config

Desktop

Programming

Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

172.31.15.10

Subnet Mask

255.255.255.0

Default Gateway

172.31.15.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address

FE80::2E0:8FFF:FE6C:EC08

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MDS

Username

Password

Top

Physical

Config

Desktop

Programming

Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

172.31.15.11

Subnet Mask

255.255.255.0

Default Gateway

172.31.15.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address

FE80::2D0:FFFF:FE8D:7E47

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MDS

Username

Password

Top

Verification

PC1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.15.1

Pinging 172.31.15.1 with 32 bytes of data:

Reply from 172.31.15.1: bytes=32 time<1ms TTL=255
Reply from 172.31.15.1: bytes=32 time<1ms TTL=255
Reply from 172.31.15.1: bytes=32 time=2ms TTL=255
Reply from 172.31.15.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.15.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

PC2:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.15.1

Pinging 172.31.15.1 with 32 bytes of data:

Reply from 172.31.15.1: bytes=32 time<1ms TTL=255
Reply from 172.31.15.1: bytes=32 time<1ms TTL=255
Reply from 172.31.15.1: bytes=32 time<1ms TTL=255
Reply from 172.31.15.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.15.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Guest PCs

The Guest PCs are assigned to the **Guest Wi-Fi VLAN (VLAN 85)**. The Guest network uses the 172.31.85.0/24 subnet, with HQ-CORE providing the default gateway through 172.31.85.1.

Device	IP Address	Subnet Mask	Default Gateway	Purpose
GUEST-PC1	172.31.85.10	255.255.255.0	172.31.85.1	Guest workstation
GUEST-PC2	172.31.85.11	255.255.255.0	172.31.85.1	Guest workstation

Guest PC Configuration

The Guest PCs are configured with static IPv4 addresses belonging to the Guest VLAN. The default gateway is set to 172.31.85.1.

Purpose:

The Guest PCs represent users connected to the Guest Wi-Fi network. The Guest network is designed to provide Internet access while preventing access to internal Finance resources. This restriction will be implemented later using ACLs.

Physical

Config

Desktop

Programming

Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

172.31.85.10

Subnet Mask

255.255.255.0

Default Gateway

172.31.85.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::2D0:58FF:FE16:1568

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MDS

Username

Password

Top

Physical

Config

Desktop

Programming

Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

172.31.85.11

Subnet Mask

255.255.255.0

Default Gateway

172.31.85.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::200:CFF:FE21:D22A

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MDS

Username

Password

Top

Verification

PC1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.85.1

Pinging 172.31.85.1 with 32 bytes of data:

Reply from 172.31.85.1: bytes=32 time=3ms TTL=255
Reply from 172.31.85.1: bytes=32 time<1ms TTL=255
Reply from 172.31.85.1: bytes=32 time<1ms TTL=255
Reply from 172.31.85.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.85.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>
```

PC2:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.85.1

Pinging 172.31.85.1 with 32 bytes of data:

Reply from 172.31.85.1: bytes=32 time=16ms TTL=255
Reply from 172.31.85.1: bytes=32 time<1ms TTL=255
Reply from 172.31.85.1: bytes=32 time<1ms TTL=255
Reply from 172.31.85.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.85.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 4ms

C:\>|
```



Plant Switch Configuration

Plant LAN Configuration

Purpose:

SW-PLANT provides Layer 2 connectivity for the Plant network. The Plant LAN uses the 172.31.10.0/24 subnet and is shared by the Plant PCs, file server, web server, PLANT-ABR and PLANT-BOUNDARY. Since this is a shared LAN for the OSPF Area 10 network, no separate VLAN is required at this stage.

Port Assignment

Switch Port	Connected Device	Purpose
Fa0/1	PLANT-ABR G0/1	OSPF Area 10 gateway
Fa0/2	PLANT-BOUNDARY G0/1	OSPF/EIGRP boundary
Fa0/3	PLANT-PC1	Plant workstation
Fa0/4	PLANT-PC2	Plant workstation
Fa0/5	FILE-SERVER	File services
Fa0/6	WEB-SERVER	Web services

Configuration

```
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fa0/3 - 6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#exit
Switch(config)#end
```

Verification

The show interfaces status command was used to verify the operational status of the Plant switch ports. The output confirms that **Fa0/1 through Fa0/6 are connected**, indicating that the router and end-device connections are operational.

```
Switch#show interfaces status
%SYS-5-CONFIG_I: Configured from console by console
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	1	a-full	a-100	10/100BaseTX
Fa0/2		connected	1	a-full	a-100	10/100BaseTX
Fa0/3		connected	1	a-full	a-100	10/100BaseTX
Fa0/4		connected	1	a-full	a-100	10/100BaseTX
Fa0/5		connected	1	a-full	a-100	10/100BaseTX
Fa0/6		connected	1	a-full	a-100	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		notconnect	1	auto	auto	10/100BaseTX
Fa0/24		notconnect	1	auto	auto	10/100BaseTX
Gig0/1		notconnect	1	auto	auto	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

Plant PC Connectivity Verification

The connectivity of the Plant workstations was verified by sending ICMP ping requests to the default gateway 172.31.10.1.

Device	Destination	Result
PLANT-PC1	172.31.10.1	4/4 replies, 0% loss
PLANT-PC2	172.31.10.1	4/4 replies, 0% loss

Verification:

PLANT-PC1 and PLANT-PC2 were tested using the ping command. Both devices successfully received replies from the Plant gateway 172.31.10.1, with **0% packet loss**. This confirms that the Plant workstations can communicate with the PLANT-ABR interface over the 172.31.10.0/24 LAN.

Screenshots

PC1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.10.1

Pinging 172.31.10.1 with 32 bytes of data:

Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

PC2:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.10.1

Pinging 172.31.10.1 with 32 bytes of data:

Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Result

The Plant workstation connectivity was successfully verified. Both Plant PCs can reach the Plant gateway, confirming that the **Plant LAN and switch connectivity are operational**.

Server Connectivity Verification

The connectivity of the Plant servers was verified by sending ICMP ping requests to the Plant gateway 172.31.10.1.

Device	Destination	Result
FILE-SERVER	172.31.10.1	4/4 replies, 0% loss
WEB-SERVER	172.31.10.1	4/4 replies, 0% loss

Verification:

Both the File Server and Web Server successfully received replies from the Plant gateway 172.31.10.1. The tests resulted in **0% packet loss**, confirming that both servers have working connectivity to the Plant LAN.

Screenshots

FILE-SERVER Gateway Ping Verification

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 172.31.10.1

Pinging 172.31.10.1 with 32 bytes of data:

Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time=2ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

WEB-SERVER Gateway Ping Verification

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 172.31.10.1

Pinging 172.31.10.1 with 32 bytes of data:

Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255
Reply from 172.31.10.1: bytes=32 time=2ms TTL=255
Reply from 172.31.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Result

The Plant LAN connectivity was successfully verified for **both workstations and servers**. All four Plant end devices can reach the Plant gateway, confirming that the Layer 2 connectivity through SW-PLANT is operational.

SW-LOOM Verification

The show interfaces status command was used to verify the operational status of the Loomcraft switch ports. The output confirms that **Fa0/1, Fa0/2 and Fa0/3 are connected**, corresponding to the Loomcraft router and the two Loomcraft PCs.

Port	Connected Device	Status
Fa0/1	LOOMCRAFT-R1 G0/1	Connected
Fa0/2	LOOM-PC1	Connected
Fa0/3	LOOM-PC2	Connected

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fa0/2 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#exit
Switch(config)#end

```

SW-LOOM Interface Status Verification

```
Switch#show interfaces status
%SYS-5-CONFIG_I: Configured from console by console

Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Fa0/1     connected   1         a-full  a-100  10/100BaseTX
Fa0/2     Fa0/2     connected   1         a-full  a-100  10/100BaseTX
Fa0/3     Fa0/3     connected   1         a-full  a-100  10/100BaseTX
Fa0/4     Fa0/4     notconnect  1         auto    auto    10/100BaseTX
Fa0/5     Fa0/5     notconnect  1         auto    auto    10/100BaseTX
Fa0/6     Fa0/6     notconnect  1         auto    auto    10/100BaseTX
Fa0/7     Fa0/7     notconnect  1         auto    auto    10/100BaseTX
Fa0/8     Fa0/8     notconnect  1         auto    auto    10/100BaseTX
Fa0/9     Fa0/9     notconnect  1         auto    auto    10/100BaseTX
Fa0/10    Fa0/10    notconnect  1         auto    auto    10/100BaseTX
Fa0/11    Fa0/11    notconnect  1         auto    auto    10/100BaseTX
Fa0/12    Fa0/12    notconnect  1         auto    auto    10/100BaseTX
Fa0/13    Fa0/13    notconnect  1         auto    auto    10/100BaseTX
Fa0/14    Fa0/14    notconnect  1         auto    auto    10/100BaseTX
Fa0/15    Fa0/15    notconnect  1         auto    auto    10/100BaseTX
Fa0/16    Fa0/16    notconnect  1         auto    auto    10/100BaseTX
Fa0/17    Fa0/17    notconnect  1         auto    auto    10/100BaseTX
Fa0/18    Fa0/18    notconnect  1         auto    auto    10/100BaseTX
Fa0/19    Fa0/19    notconnect  1         auto    auto    10/100BaseTX
Fa0/20    Fa0/20    notconnect  1         auto    auto    10/100BaseTX
Fa0/21    Fa0/21    notconnect  1         auto    auto    10/100BaseTX

Switch#
Switch#|
```

Result:

The SW-LOOM switch ports were successfully configured and verified. All three connected ports are operational.

Loomcraft PC Connectivity Verification

The connectivity of the Loomcraft workstations was verified by sending ICMP ping requests to the Loomcraft router gateway 172.31.50.1.

Device	Destination	Result
LOOM-PC1	172.31.50.1	4/4 replies, 0% loss
LOOM-PC2	172.31.50.1	4/4 replies, 0% loss

Verification:

Both Loomcraft PCs successfully received replies from 172.31.50.1 with **0% packet loss**. This confirms that the Loomcraft workstations can communicate with the LAN interface of LOOMCRAFT-R1.

LOOM-PC1 Gateway Ping Verification

```
C:\>ping 172.31.50.1

Pinging 172.31.50.1 with 32 bytes of data:

Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.50.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

LOOM-PC2 Gateway Ping Verification

```
C:\>ping 172.31.50.1

Pinging 172.31.50.1 with 32 bytes of data:

Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255
Reply from 172.31.50.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.31.50.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Result

The Loomcraft LAN connectivity was successfully verified. Both Loomcraft workstations can reach the 172.31.50.1 gateway, confirming that the **SW-LOOM Layer 2 connectivity and Loomcraft LAN addressing are operational.**

4. Static Routing

4.1 HQ-CORE Static Route

Purpose:

Before implementing dynamic routing with OSPF, a static route was configured on HQ-CORE to provide connectivity from the Headquarters network to the Plant LAN.

The Plant LAN uses the network 172.31.10.0/24, and PLANT-ABR provides the next-hop router at 172.31.0.6.

Configuration

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 172.31.10.0 255.255.255.0 172.31.0.6
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

HQ-CORE Static Route Verification

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.31.0.0/16 is variably subnetted, 9 subnets, 3 masks
C       172.31.0.0/30 is directly connected, GigabitEthernet0/0
L       172.31.0.2/32 is directly connected, GigabitEthernet0/0
C       172.31.0.4/30 is directly connected, GigabitEthernet0/1
L       172.31.0.5/32 is directly connected, GigabitEthernet0/1
S       172.31.10.0/24 [1/0] via 172.31.0.6
C       172.31.15.0/24 is directly connected, GigabitEthernet0/2.15
L       172.31.15.1/32 is directly connected, GigabitEthernet0/2.15
C       172.31.85.0/24 is directly connected, GigabitEthernet0/2.85
L       172.31.85.1/32 is directly connected, GigabitEthernet0/2.85

Router#
```

Result

The static route to the Plant LAN was successfully installed in the HQ-CORE routing table. HQ-CORE can now forward traffic destined for 172.31.10.0/24 through PLANT-ABR.

4.2 PLANT-ABR Return Routes

To provide the return path from the Plant network toward the HQ Finance and Guest networks, static routes were configured on PLANT-ABR. HQ-CORE at 172.31.0.5 was used as the next-hop router.

Configuration:

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 172.31.15.0 255.255.255.0 172.31.0.5
Router(config)#ip route 172.31.85.0 255.255.255.0 172.31.0.5
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

PLANT-ABR Static Return Route Verification

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.31.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.31.0.4/30 is directly connected, GigabitEthernet0/0
L       172.31.0.6/32 is directly connected, GigabitEthernet0/0
C       172.31.10.0/24 is directly connected, GigabitEthernet0/1
L       172.31.10.1/32 is directly connected, GigabitEthernet0/1
S       172.31.15.0/24 [1/0] via 172.31.0.5
S       172.31.85.0/24 [1/0] via 172.31.0.5

Router#
```

Result:

The return routes from the Plant network to the Finance and Guest networks were successfully configured and verified.

4.3 HQ-EDGE Default Route

Configuration:

```
Router>
Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2
Router(config)#end
```

HQ-EDGE Default Route Verification

```
Router#show ip route
%SYS-5-CONFIG_I: Configured from console by console

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 203.0.113.2 to network 0.0.0.0

    172.31.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.31.0.0/30 is directly connected, GigabitEthernet0/2
L       172.31.0.1/32 is directly connected, GigabitEthernet0/2
    203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/0
L       203.0.113.1/32 is directly connected, GigabitEthernet0/0
C       203.0.113.4/30 is directly connected, GigabitEthernet0/1
L       203.0.113.5/32 is directly connected, GigabitEthernet0/1
S*    0.0.0.0/0 [1/0] via 203.0.113.2

Router#
```

Result:

The HQ Internet edge router now has a default route toward ISP-C for destinations that are not present in its routing table.

4.4 HQ-EDGE Internal Route:

Configuration:

```
Router#
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 172.31.0.0 255.255.0.0 172.31.0.2
Router(config)#end
```

HQ-EDGE Internal Static Route Verification

```
Router#show ip route
%SYS-5-CONFIG_I: Configured from console by console

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.2 to network 0.0.0.0

    172.31.0.0/16 is variably subnetted, 3 subnets, 3 masks
S       172.31.0.0/16 [1/0] via 172.31.0.2
C       172.31.0.0/30 is directly connected, GigabitEthernet0/2
L       172.31.0.1/32 is directly connected, GigabitEthernet0/2
    203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/0
L       203.0.113.1/32 is directly connected, GigabitEthernet0/0
C       203.0.113.4/30 is directly connected, GigabitEthernet0/1
L       203.0.113.5/32 is directly connected, GigabitEthernet0/1
S*    0.0.0.0/0 [1/0] via 203.0.113.2

Router#
```

4.5 PLANT-ABR Route to HQ-EDGE:

Configuration:

```
Router#
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 172.31.0.0 255.255.255.252 172.31.0.5
Router(config)#end
```

PLANT-ABR Static Route to HQ Verification

```
Router#show ip route
%SYS-5-CONFIG_I: Configured from console by console

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    172.31.0.0/16 is variably subnetted, 7 subnets, 3 masks
S       172.31.0.0/30 [1/0] via 172.31.0.5
C       172.31.0.4/30 is directly connected, GigabitEthernet0/0
L       172.31.0.6/32 is directly connected, GigabitEthernet0/0
C       172.31.10.0/24 is directly connected, GigabitEthernet0/1
L       172.31.10.1/32 is directly connected, GigabitEthernet0/1
S       172.31.15.0/24 [1/0] via 172.31.0.5
S       172.31.85.0/24 [1/0] via 172.31.0.5

Router#
```

Verification – HQ to Plant Connectivity

```
C:\>ping 172.31.10.10

Pinging 172.31.10.10 with 32 bytes of data:

Reply from 172.31.10.10: bytes=32 time<1ms TTL=126
Reply from 172.31.10.10: bytes=32 time=1ms TTL=126
Reply from 172.31.10.10: bytes=32 time<1ms TTL=126
Reply from 172.31.10.10: bytes=32 time=1ms TTL=126

Ping statistics for 172.31.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

Successful static-routing connectivity test from PLANT-PC1 to FIN-PC1

```
C:\>ping 172.31.15.10

Pinging 172.31.15.10 with 32 bytes of data:

Reply from 172.31.15.10: bytes=32 time<1ms TTL=126
Reply from 172.31.15.10: bytes=32 time=1ms TTL=126
Reply from 172.31.15.10: bytes=32 time<1ms TTL=126
Reply from 172.31.15.10: bytes=32 time<1ms TTL=126

Ping statistics for 172.31.15.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

HQ-EDGE interface verification showing all three interfaces in up/up state.

```
Router#show ip interface brief
%SYS-5-CONFIG_I: Configured from console by console

Interface                IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0       203.0.113.1     YES manual up                  up
GigabitEthernet0/1       203.0.113.5     YES manual up                  up
GigabitEthernet0/2       172.31.0.1      YES manual up                  up
Vlan1                    unassigned      YES unset  administratively down down
Router#
```

ISP-C – ISP-Crestline

Interface Configuration

ISP-C is configured as the Crestline ISP router. Its GigabitEthernet0/0 interface is connected to the HQ-EDGE router using the 203.0.113.0/30 transit network.

Device	Interface	IP Address	Subnet Mask	Status
ISP-C	G0/0	203.0.113.2	255.255.255.252	Up/Up

Configuration:

```

Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface gigabitEthernet0/0
Router(config-if)#no ip address
Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet0/1
Router(config-if)#no ip address
Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet0/2
Router(config-if)#no ip address
Router(config-if)#exit
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 203.0.113.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 203.0.113.5 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface gigabitEthernet0/2
Router(config-if)#ip address 172.31.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end

```

ISP-C Interface Verification

```
Router#show ip interface brief
%SYS-5-CONFIG_I: Configured from console by console

Interface                IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0       203.0.113.1     YES manual up                  up
GigabitEthernet0/1       203.0.113.5     YES manual up                  up
GigabitEthernet0/2       172.31.0.1      YES manual up                  up
Vlan1                    unassigned      YES unset  administratively down down
Router#
```

Result: ISP-C's GigabitEthernet0/0 interface is correctly configured and operational.

ISP-H Verification

The show ip interface brief command was used to verify the ISP-H interface. The output confirms that GigabitEthernet0/0 is configured with 203.0.113.6 and is operational with an **up/up** status.

ISP-H Interface Verification

```
Router#show ip interface brief
%SYS-5-CONFIG_I: Configured from console by console

Interface                IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0       203.0.113.1     YES manual up                  up
GigabitEthernet0/1       203.0.113.5     YES manual up                  up
GigabitEthernet0/2       172.31.0.1      YES manual up                  up
Vlan1                    unassigned      YES unset  administratively down down
Router#
```

Result: ISP-H GigabitEthernet0/0 is correctly configured and operational.

Verification of Static Route from HQ-CORE to Plant LAN

```
Router#
Router#ping 172.31.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.31.10.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Static Routes Configured on PLANT-ABR

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.31.0.0/16 is variably subnetted, 7 subnets, 3 masks
S       172.31.0.0/30 [1/0] via 172.31.0.5
C       172.31.0.4/30 is directly connected, GigabitEthernet0/0
L       172.31.0.6/32 is directly connected, GigabitEthernet0/0
C       172.31.10.0/24 is directly connected, GigabitEthernet0/1
L       172.31.10.1/32 is directly connected, GigabitEthernet0/1
S       172.31.15.0/24 [1/0] via 172.31.0.5
S       172.31.85.0/24 [1/0] via 172.31.0.5

Router#
```

The routing table of PLANT-ABR was checked using the show ip route command. The output confirms the static routes toward the HQ network, Finance VLAN and Guest VLAN through HQ-CORE (172.31.0.5).

PLANT-ABR to Finance Gateway Connectivity Verification

```
Router#
Router#ping 172.31.15.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.31.15.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

The connectivity between PLANT-ABR and the Finance gateway was verified using ping 172.31.15.1. The test achieved **5/5 successful replies with 0% packet loss**, confirming that the static route toward the Finance network is functioning correctly.

End-to-End Connectivity Verification Using Static Routing

```
C:\>ping 172.31.10.10

Pinging 172.31.10.10 with 32 bytes of data:

Reply from 172.31.10.10: bytes=32 time<1ms TTL=126
Reply from 172.31.10.10: bytes=32 time<1ms TTL=126
Reply from 172.31.10.10: bytes=32 time<1ms TTL=126
Reply from 172.31.10.10: bytes=32 time<1ms TTL=126

Ping statistics for 172.31.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Result:

End-to-end connectivity between the Finance network and Plant LAN was verified using ping 172.31.10.10 from FIN-PC1. The test achieved 5/5 successful replies with 0% packet loss, confirming that the configured static routes provide connectivity between the two network segments.

5. OSPF Routing

5.1 OSPF Theory

Open Shortest Path First (OSPF) is a link-state dynamic routing protocol used to exchange routing information within an autonomous system. OSPF uses the Shortest Path First (SPF) algorithm to calculate the best path to a destination.

OSPF supports hierarchical network design using areas. In this project, **Area 0** is used as the backbone area between the HQ and Plant, while **Area 10** is used for the Plant LAN. This reduces the routing scope and demonstrates a multi-area OSPF design.

OSPF routers form neighbor relationships by exchanging Hello packets. After adjacency is established, routers exchange link-state information and build their routing tables.

The Plant LAN is a shared multi-access network. OSPF Designated Router (DR) and Backup Designated Router (BDR) operation will therefore be demonstrated on this network.

5.2 OSPF Area Design

Router	Interface	Network	OSPF Area
HQ-CORE	G0/0	172.31.0.0/30	Area 0
HQ-CORE	G0/1	172.31.0.4/30	Area 0
HQ-CORE	G0/2.15	172.31.15.0/24	Area 0
HQ-CORE	G0/2.85	172.31.85.0/24	Area 0
PLANT-AB R	G0/0	172.31.0.4/30	Area 0
PLANT-AB R	G0/1	172.31.10.0/24	Area 10

5.3 OSPF Configuration – HQ-CORE

OSPF process 10 was configured on HQ-CORE with router ID 1.1.1.1. The HQ-CORE interfaces belonging to the HQ-to-Plant backbone link, Finance VLAN and Guest VLAN were assigned to OSPF Area 0. The OSPF interface verification confirms that four interfaces are participating in the backbone area.

OSPF Configuration and Interface Verification on HQ-CORE

```
Router#show ip ospf
Routing Process "ospf 10" with ID 1.1.1.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 4
    Area has no authentication
    SPF algorithm executed 1 times
    Area ranges are
    Number of LSA 1. Checksum Sum 0x00cab9|
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

5.3 OSPF Configuration – PLANT-ABR

PLANT-ABR was configured with OSPF process 10 and router ID 2.2.2.2. The connection toward HQ-CORE was assigned to OSPF Area 0, while the Plant LAN was assigned to Area 10. The router was identified as an Area Border Router because it participates in both Area 0 and Area 10.

OSPF Area Configuration and Neighbor Adjacency on PLANT-ABR

```
Router#show ip ospf
Routing Process "ospf 10" with ID 2.2.2.2
Supports only single TOS(TOS0) routes
Supports opaque LSA
It is an area border router
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 2. 2 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 1
    Area has no authentication
    SPF algorithm executed 0 times
    Area ranges are
    Number of LSA 2. Checksum Sum 0x00c233
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
  Area 10
    Number of interfaces in this area is 1
    Area has no authentication
    SPF algorithm executed 1 times
--More--
21:14:07: %OSPF-5-ADJCHG: Process 10, Nbr 1.1.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
  Area ranges are
  Number of LSA 1. Checksum Sum 0x0071fb
  Number of opaque link LSA 0. Checksum Sum 0x000000
  Number of DCbitless LSA 0
  Number of indication LSA 0
  Number of DoNotAge LSA 0
  Flood list length 0

Router#
Router#show ip ospf interface brief

```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gig0/0	10	0	172.31.0.6/255.255.255.252	1	BDR	0/0	
Gig0/1	10	10	172.31.10.1/255.255.255.0	1	WAIT	0/0	

```
Router#
```

The OSPF adjacency between PLANT-ABR and HQ-CORE reached the **FULL** state, confirming successful neighbor establishment across the Area 0 backbone.

5.3 OSPF Configuration – PLANT-ABR / HQ-CORE Neighbor Verification

OSPF Neighbor Adjacency Between HQ-CORE and PLANT-ABR

```
Router#
Router#show ip ospf interface brief

```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gig0/0	10	0	172.31.0.2/255.255.255.252	1	WAIT	0/0	
Gig0/1	10	0	172.31.0.5/255.255.255.252	1	WAIT	0/0	
Gig	10	0	172.31.15.1/255.255.255.0	1	WAIT	0/0	
Gig	10	0	172.31.85.1/255.255.255.0	1	WAIT	0/0	

```
Router#
Router#
Router#
Router#
Router#
Router#
21:14:02: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

Router#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/BDR	00:00:33	172.31.0.6	GigabitEthernet0/1

```
Router#
```

The OSPF neighbor relationship between HQ-CORE and PLANT-ABR was successfully established. The show ip ospf neighbor command on HQ-CORE displays PLANT-ABR with Router ID 2.2.2.2 in the FULL state, confirming successful

OSPF adjacency across Area 0.

Configuration and Neighbor Adjacency on PLANT-BOUNDARY

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 10
Router(config-router)#router-id 3.3.3.3
Router(config-router)#network 172.31.10.0 0.0.0.255 area 10
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf
Routing Process "ospf 10" with ID 3.3.3.3
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area 10
    Number of interfaces in this area is 1
    Area has no authentication
    SPF algorithm executed 1 times
    Area ranges are
    Number of LSA 1. Checksum Sum 0x002244
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

Result:

PLANT-BOUNDARY was successfully configured with OSPF process 10 and Router ID 3.3.3.3. Its Plant LAN interface was assigned to OSPF Area 10. The OSPF adjacency with PLANT-ABR reached the FULL state, confirming successful neighbor establishment within Area 10.

OSPF Area 10 Neighbor Adjacency on PLANT-ABR

```
Router#
21:36:17: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
show ip ospf interface brief

```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gig0/1	10	10	172.31.10.2/255.255.255.0	1	BDR	0/0	

```
Router#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/DR	00:00:34	172.31.10.1	GigabitEthernet0/1

```
Router#
```

Result:

The show ip ospf neighbor command on PLANT-ABR confirms a FULL OSPF adjacency with PLANT-BOUNDARY. PLANT-ABR is identified as the DR and PLANT-BOUNDARY as the BDR on the shared Plant LAN, demonstrating OSPF neighbor formation and DR/BDR operation in Area 10.

OSPF Inter-Area Route Learned by HQ-CORE

```
Router#show ip route ospf
    172.31.0.0/16 is variably subnetted, 9 subnets, 3 masks
O IA    172.31.10.0 [110/2] via 172.31.0.6, 00:26:34, GigabitEthernet0/1

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    172.31.0.0/16 is variably subnetted, 9 subnets, 3 masks
C        172.31.0.0/30 is directly connected, GigabitEthernet0/0
L        172.31.0.2/32 is directly connected, GigabitEthernet0/0
C        172.31.0.4/30 is directly connected, GigabitEthernet0/1
L        172.31.0.5/32 is directly connected, GigabitEthernet0/1
O IA    172.31.10.0/24 [110/2] via 172.31.0.6, 00:26:39, GigabitEthernet0/1
C        172.31.15.0/24 is directly connected, GigabitEthernet0/2.15
L        172.31.15.1/32 is directly connected, GigabitEthernet0/2.15
C        172.31.85.0/24 is directly connected, GigabitEthernet0/2.85
L        172.31.85.1/32 is directly connected, GigabitEthernet0/2.85

Router#
```

Result:

After removing the initial static route for the Plant LAN, HQ-CORE dynamically learned the 172.31.10.0/24

network through OSPF. The route appears as an OSPF inter-area (O IA) route via PLANT-ABR at 172.31.0.6, confirming successful routing between OSPF Area 0 and Area 10.

Successful Connectivity to Plant Network Through OSPF

```
Router>ping 172.31.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.31.10.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router>ping 172.31.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.31.10.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router>
```

Result:

Connectivity from HQ-CORE to the Plant network was successfully verified. Both the PLANT-ABR interface (172.31.10.1) and PLANT-PC1 (172.31.10.10) responded successfully with 100% packet delivery, confirming end-to-end communication through the OSPF-routed network.

6. EIGRP Routing

6.1 EIGRP Theory

Enhanced Interior Gateway Routing Protocol (EIGRP) is a dynamic routing protocol used to exchange routing information within an autonomous system. EIGRP uses the Diffusing Update Algorithm (DUAL) to determine loop-free paths and maintain backup routes when available.

In this project, Loomcraft Fabrics continues to use EIGRP with Autonomous System (AS) number 950. The existing Loomcraft routing domain is maintained, while EIGRP will later be connected to the OSPF domain through PLANT-BOUNDARY using route redistribution.

6.2 EIGRP Network Design

Device	Interface	Network	EIGRP AS
PLANT-BOUNDARY	G0/0	172.31.20.0/30	950
LOOMCRAFT-R1	G0/0	172.31.20.0/30	950
LOOMCRAFT-R1	G0/1	172.31.50.0/24	950

6.3 EIGRP Configuration – LOOMCRAFT-R1

LOOMCRAFT-R1 was configured to continue using EIGRP Autonomous System 950. The router advertises the point-to-point network 172.31.20.0/30 toward PLANT-BOUNDARY and the Loomcraft LAN 172.31.50.0/24.

Configuration

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router eigrp 950
Router(config-router)#network 172.31.20.0 0.0.0.3
Router(config-router)#network 172.31.50.0 0.0.0.255
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 950

Router#show ip route eigrp

Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       172.31.20.2     YES manual up          up
GigabitEthernet0/1       172.31.50.1     YES manual up          up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down

Router#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce    Holdtme    Capability    Platform    Port ID
Router       Gig 0/0          153        R              C2900       Gig 0/0
Switch       Gig 0/1          153        S              2960        Fas 0/1
Router#
```

6.4 EIGRP Configuration – PLANT-BOUNDARY

PLANT-BOUNDARY was configured with EIGRP Autonomous System 950 on the Loomcraft-facing network. The 172.31.20.0/30 network was included in EIGRP to establish the routing relationship with LOOMCRAFT-R1.

EIGRP AS 950 Configuration on PLANT-BOUNDARY

```
Router#
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no ip address
Router(config-if)#ip address 172.31.20.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
22:03:17: %OSPF-5-ADJCHG: Process 10, Nbr 1.1.1.1 on GigabitEthernet0/0 from FULL
Router#
```

6.5 EIGRP Neighbor Verification

The EIGRP neighbor relationship between PLANT-BOUNDARY and LOOMCRAFT-R1 was successfully established using EIGRP Autonomous System 950. The show ip eigrp neighbors command on PLANT-BOUNDARY displays LOOMCRAFT-R1 at 172.31.20.2 through GigabitEthernet0/0, confirming successful EIGRP adjacency.

EIGRP Neighbor Adjacency Between PLANT-BOUNDARY and LOOMCRAFT-R1

```
Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 950
H   Address          Interface      Hold Uptime    SRTT   RTO   Q   Seq
                   (sec)                  (ms)                Cnt   Num
0   172.31.20.2       Gig0/0         11  00:00:07    40    1000   0    1

Router#
```

Result

The successful EIGRP adjacency confirms that the Loomcraft routing domain is communicating with PLANT-BOUNDARY using EIGRP AS 950.

6.6 EIGRP Route Verification

The EIGRP routing table on PLANT-BOUNDARY was verified using the `show ip route eigrp` command. The router learned the Loomcraft LAN 172.31.50.0/24 through LOOMCRAFT-R1 at 172.31.20.2. The route is identified by the D code, confirming that the network was learned through EIGRP.

EIGRP Route to the Loomcraft LAN on PLANT-BOUNDARY

```
Router#  
Router#show ip route eigrp  
    172.31.0.0/16 is variably subnetted, 5 subnets, 3 masks  
D       172.31.50.0/24 [90/28416] via 172.31.20.2, 00:02:20, GigabitEthernet0/0  
Router#
```

Result:

The successful EIGRP route confirms that PLANT-BOUNDARY can dynamically learn the Loomcraft LAN through EIGRP AS 950.

6.7 EIGRP-to-OSPF Route Redistribution

Route redistribution was configured on PLANT-BOUNDARY to redistribute EIGRP AS 950 routes into OSPF process 10. The `redistribute eigrp 950 subnets` command was used to allow EIGRP networks to be advertised into the OSPF domain. The OSPF database shows Type-5 AS External Link States for the redistributed networks, confirming that the redistribution process is active.

EIGRP-to-OSPF Route Redistribution on PLANT-BOUNDARY

```
Router#show ip route ospf
Router#
Router#show ip ospf database
      OSPF Router with ID (3.3.3.3) (Process ID 10)

      Router Link States (Area 10)

Link ID      ADV Router    Age      Seq#          Checksum Link count
2.2.2.2      2.2.2.2        444      0x8000000a   0x00d5b6 1
3.3.3.3      3.3.3.3        14       0x8000000b   0x0098e8 1

      Net Link States (Area 10)

Link ID      ADV Router    Age      Seq#          Checksum
172.31.10.2  3.3.3.3       464      0x80000002   0x001152

      Type-5 AS External Link States

Link ID      ADV Router    Age      Seq#          Checksum Tag
172.31.20.0  3.3.3.3       14       0x80000001   0x004691 0
172.31.50.0  3.3.3.3       14       0x80000001   0x000da9 0
```

Result

The OSPF database on PLANT-BOUNDARY contains Type-5 AS External Link States for 172.31.50.0/24 and 172.31.20.0/30, confirming that EIGRP routes are being redistributed into OSPF.

6.7 EIGRP-to-OSPF Route Redistribution, after the existing PLANT-BOUNDARY verification.

The redistributed Loomcraft network was further verified on PLANT-ABR using the show ip route 172.31.50.0 command. PLANT-ABR learned the 172.31.50.0/24 network through OSPF from PLANT-BOUNDARY (172.31.10.2). The OSPF database also contains a Type-5 AS External Link State for the Loomcraft network, confirming that the redistributed EIGRP route has reached the OSPF Area 10 router.

Redistributed Loomcraft Route Learned by PLANT-ABR

```
Router>show ip route 172.31.50.0
Routing entry for 172.31.50.0/24
Known via "ospf 10", distance 110, metric 20, type intra area
  Last update from 172.31.10.2 on GigabitEthernet0/1, 00:06:11 ago
  Routing Descriptor Blocks:
    * 172.31.10.2, from 3.3.3.3, 00:06:11 ago, via GigabitEthernet0/1
      Route metric is 20, traffic share count is 1

Router>show ip ospf database
        OSPF Router with ID (2.2.2.2) (Process ID 10)

                Router Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum Link count
2.2.2.2      2.2.2.2        764      0x80000008    0x00f953  0

                Summary Net Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum
172.31.10.0  2.2.2.2        51       0x80000006    0x00a4d6

                Summary ASB Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum
3.3.3.3      2.2.2.2        376      0x80000005    0x00b68e

                Router Link States (Area 10)

Link ID      ADV Router    Age      Seq#          Checksum Link count
2.2.2.2      2.2.2.2        811      0x8000000a    0x00d5b6  1
3.3.3.3      3.3.3.3        382      0x8000000b    0x0098e8  1

                Net Link States (Area 10)

Link ID      ADV Router    Age      Seq#          Checksum
172.31.10.2  3.3.3.3        832      0x80000002    0x001152

                Type-5 AS External Link States

Link ID      ADV Router    Age      Seq#          Checksum Tag
172.31.20.0  3.3.3.3        382      0x80000001    0x004691  0
172.31.50.0  3.3.3.3        382      0x80000001    0x000da9  0
Router>
```

Result

The successful route installation on PLANT-ABR confirms that the EIGRP-to-OSPF redistribution is propagating the Loomcraft network beyond PLANT-BOUNDARY.

The redistributed Loomcraft network was verified from HQ-CORE using the show ip route 172.31.50.0 command. HQ-CORE successfully learned the 172.31.50.0/24 network through OSPF, with PLANT-ABR (172.31.0.6) as the next hop. The OSPF neighbor table also confirmed a FULL adjacency with PLANT-ABR. This verifies that the Loomcraft network is successfully propagated from the EIGRP domain through the OSPF domain to the HQ.

HQ-CORE Verification of the Redistributed Loomcraft Network

```
Router#
Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          1     FULL/BDR        00:00:36    172.31.0.6   GigabitEthernet0/1
Router#show ip route 172.31.50.0
Routing entry for 172.31.50.0/24
Known via "ospf 10", distance 110, metric 20, type intra area
  Last update from 172.31.0.6 on GigabitEthernet0/1, 00:00:31 ago
  Routing Descriptor Blocks:
    * 172.31.0.6, from 2.2.2.2, 00:00:31 ago, via GigabitEthernet0/1
      Route metric is 20, traffic share count is 1

Router#
```

Result

The successful route installation on HQ-CORE confirms end-to-end propagation of the Loomcraft network through the EIGRP-to-OSPF routing boundary.

OSPF routes were redistributed into EIGRP on PLANT-BOUNDARY and verified from LOOMCRAFT-R1. The routing table shows multiple EIGRP external (D EX) routes, including the HQ and Plant networks, learned through PLANT-BOUNDARY at 172.31.20.1. This confirms that OSPF routes are successfully being redistributed into EIGRP AS 950.

OSPF Routes Redistributed into EIGRP on LOOMCRAFT-R1

```
Router>show ip route eigrp
      172.31.0.0/16 is variably subnetted, 9 subnets, 3 masks
D EX   172.31.0.0/30 [170/281856] via 172.31.20.1, 00:00:07, GigabitEthernet0/0
D EX   172.31.0.4/30 [170/281856] via 172.31.20.1, 00:00:07, GigabitEthernet0/0
D EX   172.31.10.0/24 [170/281856] via 172.31.20.1, 00:00:07, GigabitEthernet0/0
D EX   172.31.15.0/24 [170/281856] via 172.31.20.1, 00:00:07, GigabitEthernet0/0
D EX   172.31.85.0/24 [170/281856] via 172.31.20.1, 00:00:07, GigabitEthernet0/0

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.31.0.0/16 is variably subnetted, 9 subnets, 3 masks
D EX   172.31.0.0/30 [170/281856] via 172.31.20.1, 00:00:11, GigabitEthernet0/0
D EX   172.31.0.4/30 [170/281856] via 172.31.20.1, 00:00:11, GigabitEthernet0/0
D EX   172.31.10.0/24 [170/281856] via 172.31.20.1, 00:00:11, GigabitEthernet0/0
D EX   172.31.15.0/24 [170/281856] via 172.31.20.1, 00:00:11, GigabitEthernet0/0
C       172.31.20.0/30 is directly connected, GigabitEthernet0/0
L       172.31.20.2/32 is directly connected, GigabitEthernet0/0
C       172.31.50.0/24 is directly connected, GigabitEthernet0/1
L       172.31.50.1/32 is directly connected, GigabitEthernet0/1
D EX   172.31.85.0/24 [170/281856] via 172.31.20.1, 00:00:11, GigabitEthernet0/0

Router>
```

Result

The presence of EIGRP external routes on LOOMCRAFT-R1 confirms successful bidirectional route redistribution between the OSPF and EIGRP routing domains.

End-to-end connectivity was verified from LOOMCRAFT-R1 to the Plant and Finance networks. The router successfully reached PLANT-PC1 at 172.31.10.10 and FIN-PC1 at 172.31.15.10, with the final tests achieving 100% success (5/5 packets). This confirms successful bidirectional routing between the EIGRP-based Loomcraft network and the OSPF-based internal networks.

End-to-End Connectivity Verification from Loomcraft

```
Router>ping 172.31.15.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.31.15.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router>ping 172.31.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.31.10.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router>
```

Result:

Successful ping tests confirm that routes redistributed between OSPF and EIGRP provide end-to-end connectivity between Loomcraft, Plant and Finance networks.

EIGRP Route Summarization

Configuration:

The PLANT-BOUNDARY router was configured to summarize the internal Westfold networks before advertising them toward Loomcraft using EIGRP AS 950. The summary address 172.31.0.0/16 was configured on the interface connecting PLANT-BOUNDARY to LOOMCRAFT-R1.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router eigrp 950
Router(config-router)#redistribute ospf 10 metric 10000 100 255 1 1500
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip summary-address eigrp 950 172.31.0.0 255.255.0.0
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
%DUAL-5-NBRCHANGE: IP-EIGRP 950: Neighbor 172.31.20.2 (GigabitEthernet0/0) is up: new adjacency

Router#
```

EIGRP Route Summarization Verified on LOOMCRAFT-R1

```
Router>
Router>show ip route eigrp
      172.31.0.0/16 is variably subnetted, 5 subnets, 4 masks
D       172.31.0.0/16 [90/3072] via 172.31.20.1, 00:00:08, GigabitEthernet0/0

Router>
```

Verification of Directly Connected Plant LAN on PLANT-BOUNDARY

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/1
Router(config-if)#no ip address
Router(config-if)#ip address 172.31.10.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
00:16:05: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/1 from FULL to DOWN, Neighbor
00:16:05: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/1 from LOADING to FULL, Loading
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route 172.31.10.0
Routing entry for 172.31.10.0/24
Known via "connected", distance 0, metric 0 (connected, via interface)
  Redistributing via eigrp 950
  Routing Descriptor Blocks:
    * directly connected, via GigabitEthernet0/1
      Route metric is 0, traffic share count is 1

Router#
```

The Plant LAN was verified on PLANT-BOUNDARY using the show ip route 172.31.10.0 command. The routing table correctly identifies 172.31.10.0/24 as a directly connected network through GigabitEthernet0/1. This confirms that the Plant LAN is correctly connected to PLANT-BOUNDARY and is no longer incorrectly represented as an EIGRP-learned route.

Result:

The verification confirms that the Plant LAN routing entry has been restored to the correct connected state.

End-to-End Connectivity Verification from Loomcraft to Plant

```
Router>
Router>ping 172.31.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.31.10.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router>ping 172.31.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.31.10.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router>
```

End-to-end connectivity from the Loomcraft EIGRP network to the Plant network was verified from LOOMCRAFT-R1. The router successfully reached the Plant gateway at 172.31.10.1 and PLANT-PC1 at 172.31.10.10. Both tests achieved a 100% success rate with 5 out of 5 packets received.

Result:

The successful ping tests confirm that the Plant network is reachable from the Loomcraft EIGRP domain through the configured OSPF and EIGRP routing paths.

EIGRP Route Filtering and Connectivity Verification

```
Router>enable
Router#show ip route eigrp
    172.31.0.0/16 is variably subnetted, 5 subnets, 4 masks
D       172.31.0.0/16 [90/3072] via 172.31.20.1, 4294967273:4294967266:4294967280, GigabitEthernet0/0

Router#
```

OSPF-to-EIGRP redistribution was configured on PLANT-BOUNDARY using the match internal option. This restricts redistribution to internal OSPF routes. The configuration was verified on LOOMCRAFT-R1, where the summarized Westfold internal network 172.31.0.0/16 was

learned through EIGRP via 172.31.20.1. Connectivity to the Plant gateway and PLANT-PC1 was also successfully verified with 100% packet delivery.

Result:

The verification confirms that route-type filtering is active while maintaining connectivity between the Loomcraft EIGRP domain and the internal Westfold networks.

7. BGP Routing

7.1 BGP Theory

Border Gateway Protocol (BGP) is a path-vector routing protocol used to exchange routing information between different Autonomous Systems (AS). In this project, BGP is used to connect the Westfold Textile Mills network to two Internet Service Providers, ISP-Crestline and ISP-Harbor.

External BGP (eBGP) is used because the routers belong to different Autonomous Systems. HQ-EDGE belongs to AS 65120, ISP-Crestline belongs to AS 65022, and ISP-Harbor belongs to AS 65023.

BGP allows the network to exchange routes with both ISPs and provides redundant Internet connectivity. BGP path-selection attributes such as Local Preference and AS-Path are used to influence route selection.

Local Preference is used within an Autonomous System to influence the preferred outbound path. A higher Local Preference value is preferred.

AS-Path records the Autonomous Systems through which a BGP route has passed. AS-Path manipulation can be used to influence the preferred return path.

7.2 BGP AS Design

Router	Role	BGP AS
HQ-EDGE	Westfold Internet Edge	65120
ISP-C	ISP-Crestline	65022
ISP-H	ISP-Harbor	65023

The eBGP connections use the following transit networks:

Connection	Network
HQ-EDGE ↔ ISP-C	203.0.113.0/30
HQ-EDGE ↔ ISP-H	203.0.113.4/30

7.3 BGP Configuration – ISP-Crestline

ISP-Crestline was configured with BGP Autonomous System 65022. An external BGP neighbor was configured with HQ-EDGE at 203.0.113.1, which belongs to AS 65120.

Configuration

```
Router#
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65022
Router(config-router)#neighbor 203.0.113.1 remote-as 65120
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

6.4 BGP Configuration – HQ-EDGE

HQ-EDGE was configured with BGP Autonomous System 65120. Two external BGP neighbors were configured for the redundant ISP connections.

Configuration

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65023
Router(config-router)#neighbor 203.0.113.5 remote-as 65120
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Verification

The show ip bgp summary command was used to verify the BGP neighbor relationships.

BGP Dual-ISP Neighbor Adjacency on HQ-EDGE

```
Router#show ip bgp summary
BGP router identifier 203.0.113.5, local AS number 65120
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ  OutQ Up/Down  State/PfxRcd
203.0.113.2    4 65022      0       0        1    0    0 14:46:28        4
203.0.113.6    4 65023      0       0        1    0    0 14:46:28        4

Router#
```

Result:

Both ISP connections were successfully configured as external BGP neighbors, establishing the foundation for redundant Internet routing.

7.5 BGP Route Advertisement

The public network 198.51.100.24/29 is used for the external addressing and NAT requirements of the network. The network is advertised through BGP from HQ-EDGE.

Public Network Advertised through BGP

```
Router#
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 198.51.100.24 255.255.255.248 Null0
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65120
Router(config-router)#network 198.51.100.24 mask 255.255.255.248
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification:

The public network 198.51.100.24/29 was advertised from HQ-EDGE using BGP. The show ip bgp command confirms that the network appears in the BGP routing table with the *> status, indicating that the route is valid and selected as the best route. The route was originated locally by HQ-EDGE using the configured BGP network statement.

Public Network Advertisement in the BGP Routing Table

```
Router#show ip bgp
BGP table version is 2, local router ID is 10.10.10.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop              Metric LocPrf Weight Path
*> 198.51.100.24/29  0.0.0.0                      0      0 32768 i

Router#|
```

Result:

The public network was successfully installed in the BGP routing table for advertisement to the connected ISP networks.

7.6 BGP Local Preference

Local Preference was configured to influence the preferred outbound path toward the Internet. A higher Local Preference value is preferred within the Westfold AS.

The BGP configuration on HQ-EDGE was verified using the `show ip bgp summary` command. The router is operating under Autonomous System 65120 and has established eBGP sessions with ISP-Crestline (AS 65022) and ISP-Harbor (AS 65023). Both neighbors are in the established state and are receiving BGP prefixes, confirming successful dual-ISP BGP connectivity.

Figure 26: BGP Local Preference Verification

```
Router#show ip bgp summary
BGP router identifier 10.10.10.1, local AS number 65120
BGP table version is 2, main routing table version 6
1 network entries using 132 bytes of memory
1 path entries using 52 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 240 total bytes of memory
BGP activity 1/0 prefixes, 1/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent   TblVer   InQ  OutQ Up/Down  State/PfxRcd
203.0.113.2    4 65022      0       0        2    0    0 15:04:02        4
203.0.113.6    4 65023      0       0        2    0    0 15:04:02        4

Router#
```

Result:

The BGP output was used to verify the configured Local Preference and the resulting path selection.

7.7 BGP AS-Path Manipulation

AS-Path manipulation is a BGP technique used to influence route selection by modifying the AS-Path attribute. AS-Path prepending adds additional occurrences of an Autonomous System number to a route's path, making the path appear longer to other BGP routers. A longer AS-Path is generally less preferred when BGP compares otherwise suitable routes.

In the Packet Tracer IOS used for this project, the commands required for AS-Path manipulation, including route-map based configuration, were not available. Therefore, AS-Path prepending was not implemented. The dual-ISP BGP configuration and neighbor establishment were successfully implemented and verified.

7.8 Final BGP Verification

Final BGP Routing and Neighbor Verification

Final BGP verification was performed on HQ-EDGE using the `show ip bgp summary` and `show ip bgp commands`. The BGP summary confirms active eBGP sessions with ISP-Crestline (AS 65022) and ISP-Harbor (AS 65023). The BGP routing table also contains the public network 198.51.100.24/29 with the `*>` status, indicating that the route is valid and selected as the best route.

```
Router#show ip bgp
BGP table version is 2, local router ID is 10.10.10.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop              Metric LocPrf Weight Path
*> 198.51.100.24/29  0.0.0.0                      0      0 32768 i

Router#
```

Result:

The final verification confirms the operation of the dual-ISP BGP configuration and the exchange of the configured routes.

7.9 BGP Failover Verification

The redundant ISP connections were tested by temporarily making one ISP path unavailable and verifying that connectivity could continue through the remaining ISP connection.

BGP failover was tested by temporarily shutting down the HQ-EDGE interface connected to ISP-Harbor. During the test, the ISP-Harbor connection was administratively disabled while the ISP-Crestline connection remained operational. The affected interface was subsequently restored using the no shutdown command. Final BGP verification using show ip bgp summary confirmed that both ISP-Crestline (AS 65022) and ISP-Harbor (AS 65023) were again active BGP neighbors.

BGP Dual-ISP Failover Verification

```
Router#show ip bgp summary
BGP router identifier 10.10.10.1, local AS number 65120
BGP table version is 2, main routing table version 6
1 network entries using 132 bytes of memory
1 path entries using 52 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 240 total bytes of memory
BGP activity 1/0 prefixes, 1/0 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
203.0.113.2	4	65022	0	0	2	0	0	15:13:02	4
203.0.113.6	4	65023	0	0	2	0	0	15:13:02	4

Result:

The failover test verifies the redundant Internet routing design and confirms that the remaining ISP connection can be used when one ISP path becomes unavailable.

8. NAT AND PAT

8.1 NAT/PAT Theory

Network Address Translation (NAT) is used to translate private IP addresses into public IP addresses when internal devices communicate with external networks. NAT allows devices using private IP addresses to access external networks without directly exposing their private addresses.

Port Address Translation (PAT), also known as NAT overload, allows multiple internal devices to share public IP addresses by using different port numbers to identify individual connections.

In this project, the internal Westfold network uses the private address space 172.31.0.0/16, while the public block 198.51.100.24/29 is reserved for NAT/PAT. HQ-EDGE performs NAT and PAT because it provides connectivity between the internal network and the Internet Service Providers.

Static NAT provides a fixed one-to-one mapping between an internal private IP address and a public IP address. In this project, Static NAT is used to provide external access to the internal Web Server.

8.2 NAT Address Design

Purpose	Address
Internal Network	172.31.0.0/16
Public NAT Block	198.51.100.24/29
Public Usable Range	198.51.100.25 – 198.51.100.30

HQ-EDGE Loopback	198.51.100.25
PAT Pool	198.51.100.26 – 198.51.100.29
Static NAT Web Server	198.51.100.30
Web Server	172.31.10.21

The public block 198.51.100.24/29 provides six usable addresses. The existing HQ-EDGE Loopback uses 198.51.100.25, so the remaining addresses are allocated for PAT and Static NAT.

8.3 NAT Interface Configuration

HQ-EDGE acts as the NAT router. The interface connected toward the internal network is configured as the NAT inside interface, while the ISP-facing interfaces are configured as NAT outside interfaces.

Configuration

```

Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

```
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/2
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
Router#
```

8.4 PAT Configuration

A standard access list is used to identify the internal 172.31.0.0/16 network. A public address pool is then configured and PAT overload is enabled so that multiple internal devices can share the available public addresses.

Configuration

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#access-list 1 permit 172.31.0.0 0.0.255.255
Router(config)#
Router(config)#ip nat pool WESTFOLD-PAT 198.51.100.26 198.51.100.29 netmask 255.255.255.248
Router(config)#
Router(config)#ip nat inside source list 1 pool WESTFOLD-PAT overload
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

8.5 PAT Verification

PAT was verified using the show ip nat translations and show ip nat statistics commands after generating traffic from internal devices toward the external network.

Verification

The NAT translation table should display dynamically created translations between internal private addresses and the configured public PAT addresses.

```
Router>show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 198.51.100.26:1024 172.31.15.10:32   203.0.113.2:32     203.0.113.2:1024
icmp 198.51.100.26:29   172.31.15.10:29   203.0.113.2:29     203.0.113.2:29
icmp 198.51.100.26:30   172.31.15.10:30   203.0.113.2:30     203.0.113.2:30
icmp 198.51.100.26:31   172.31.15.10:31   203.0.113.2:31     203.0.113.2:31
icmp 198.51.100.26:32   172.31.0.2:32     203.0.113.2:32     203.0.113.2:32
icmp 198.51.100.26:33   172.31.0.2:33     203.0.113.2:33     203.0.113.2:33
icmp 198.51.100.26:34   172.31.0.2:34     203.0.113.2:34     203.0.113.2:34
icmp 198.51.100.26:35   172.31.0.2:35     203.0.113.2:35     203.0.113.2:35
icmp 198.51.100.26:36   172.31.0.2:36     203.0.113.2:36     203.0.113.2:36
icmp 198.51.100.26:37   172.31.0.2:37     203.0.113.2:37     203.0.113.2:37
icmp 198.51.100.26:38   172.31.0.2:38     203.0.113.2:38     203.0.113.2:38
icmp 198.51.100.26:39   172.31.0.2:39     203.0.113.2:39     203.0.113.2:39
icmp 198.51.100.26:40   172.31.0.2:40     203.0.113.2:40     203.0.113.2:40

Router>
```

Result

The successful NAT translations confirm that internal private addresses can be translated to public addresses using PAT.

PAT Translation Verification on HQ-EDGE

```
Router>ping 203.0.113.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router>
```

```
C:\>ping 203.0.113.2

Pinging 203.0.113.2 with 32 bytes of data:

Reply from 203.0.113.2: bytes=32 time<1ms TTL=253
Reply from 203.0.113.2: bytes=32 time=1ms TTL=253
Reply from 203.0.113.2: bytes=32 time<1ms TTL=253
Reply from 203.0.113.2: bytes=32 time<1ms TTL=253

Ping statistics for 203.0.113.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

8.6 Static NAT Configuration

Static NAT is configured to provide external access to the internal Web Server. The Web Server uses the private address 172.31.10.21 and is mapped to the public address 198.51.100.30.

Configuration

Static NAT Translation for the Web Server

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip nat inside source static 172.31.10.21 198.51.100.30
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

8.7 Static NAT Verification

The Static NAT mapping is verified using the NAT translation table.

Verification Command

Static NAT Translation for the Web Server

```
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
---  198.51.100.30        172.31.10.21     ---               ---
Router#
```

Result

The Static NAT translation confirms that the internal Web Server has been assigned a fixed public IP address for external access.

8.8 External Web Server Access Verification

The public Static NAT address is tested from an external network to verify that the translated address can reach the internal Web Server.

Verification

The Web Server can also be accessed using its translated public address through the Packet Tracer web browser if the simulated path supports HTTP access.

External Access to Web Server Through Static NAT

```
Router#ping 198.51.100.30
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.51.100.30, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
Router#w
```

Result

Successful external connectivity confirms that the public Static NAT address correctly forwards traffic to the internal Web Server.

8.9 Final NAT Verification

The final NAT configuration is verified on HQ-EDGE using the NAT translation and statistics commands.

Final NAT and PAT Verification on HQ-EDGE

```
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
---  198.51.100.30       172.31.10.21      ---               ---

Router#
%BGP-3-NOTIFICATION: received from neighbor 203.0.113.2 4/0 (hold time expired) 0 bytes
%BGP-5-ADJCHANGE: neighbor 203.0.113.2 Up

Router#
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 198.51.100.30:31  172.31.10.21:31  203.0.113.2:31    203.0.113.2:31
icmp 198.51.100.30:32  172.31.10.21:32  203.0.113.2:32    203.0.113.2:32
icmp 198.51.100.30:33  172.31.10.21:33  203.0.113.2:33    203.0.113.2:33
icmp 198.51.100.30:34  172.31.10.21:34  203.0.113.2:34    203.0.113.2:34
icmp 198.51.100.30:35  172.31.10.21:35  203.0.113.2:35    203.0.113.2:35
---  198.51.100.30       172.31.10.21      ---               ---

Router#
```

Result

The final verification confirms successful implementation of PAT for internal network access and Static NAT for external access to the Web Server.

9. ACCESS CONTROL LISTS (ACL)

9.1 ACL Theory

An **Access Control List (ACL)** is a set of rules used on a router or Layer 3 device to control network traffic. ACLs examine packets based on specified conditions and either **permit** or **deny** the traffic.

ACLs are used to improve network security by controlling which devices or networks are allowed to communicate with other networks. In this project, ACLs are used to restrict communication between the **Guest VLAN** and the **Finance VLAN** while allowing other required network communication.

The Westfold Textile Mills network uses separate VLANs for Finance and Guest users. The Finance network uses the subnet 172.31.15.0/24, while the Guest network uses 172.31.85.0/24. The

separation of these networks allows an ACL to control communication between them.

The main ACL requirement in this project is to **prevent Guest users from accessing Finance resources**. This protects Finance systems from unauthorized access while maintaining connectivity to other permitted networks. The case study specifically identifies ACLs as the security mechanism for restricting Guest access to the Finance network.

9.2 Types of ACLs

ACLs used in Cisco networks can generally be classified into **Standard ACLs** and **Extended ACLs**.

Standard ACL

A **Standard ACL** filters traffic mainly based on the **source IP address**.

It can be used when the requirement is to allow or deny traffic from a particular source network or host.

Example:

```
access-list 10 deny 172.31.85.0 0.0.0.255
```

This rule denies traffic originating from the Guest network.

Extended ACL

An **Extended ACL** provides more detailed traffic control. It can filter traffic based on:

- Source IP address
- Destination IP address
- Protocol
- Port number

Extended ACLs are useful when a specific type of communication between two networks needs to be controlled.

For example, an extended ACL can specifically deny Guest traffic destined for the Finance network while allowing other traffic.

9.3 ACL Processing

ACL rules are processed **from top to bottom**.

The router checks each packet against the ACL entries in sequence. Once a packet matches a rule, the corresponding action is applied.

The two basic actions are:

- **Permit** — allows the traffic.
- **Deny** — blocks the traffic.

If no configured ACL entry matches a packet, Cisco ACLs apply an implicit **deny** at the end of the ACL. Therefore, ACL rules must be designed carefully so that required traffic is not unintentionally blocked.

9.4 ACL Placement

ACLs can be applied to router interfaces in either the **inbound** or **outbound** direction.

- **Inbound (in)** — traffic is checked as it enters the interface.
- **Outbound (out)** — traffic is checked as it leaves the interface.

For this project, the ACL will be placed at the routing point where traffic between the **Guest VLAN (172.31.85.0/24)** and **Finance VLAN (172.31.15.0/24)** can be controlled.

9.5 ACL Design for Westfold Textile Mills

The ACL security requirement is based on the separation of the Finance and Guest networks.

Network	Subnet	Purpose
Finance VLAN	172.31.15.0/24	Finance users and systems
Guest VLAN	172.31.85.0/24	Guest users
Finance Gateway	172.31.15.1	HQ-CORE
Guest Gateway	172.31.85.1	HQ-CORE

The ACL will be configured to **restrict Guest access to the Finance network** while allowing the required communication to other permitted networks.

This supports the case-study requirement to protect Finance systems from Guest users.

9.6 Purpose of ACL in This Project

The ACL implementation provides the following security control:

1. Guest users are prevented from accessing the Finance network.
2. Finance systems remain protected from unauthorized Guest traffic.
3. Other required network communication can continue.

4. Network traffic can be controlled at the routing point using Cisco IOS ACL rules.
5. ACL verification can be performed using connectivity tests and Cisco IOS commands.

The ACL configuration therefore adds a layer of **network access control** between the Finance and Guest VLANs without changing the underlying routing design.

9.7 ACL Configuration

The ACL is configured on **HQ-CORE**, which performs inter-VLAN routing for the Finance and Guest networks.

The Finance VLAN uses the subnet 172.31.15.0/24, while the Guest VLAN uses 172.31.85.0/24. The ACL is designed to prevent Guest users from initiating communication with the Finance network while allowing other required traffic.

ACL Configuration

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#ip access-list extended GUEST-FINANCE-BLOCK
Router(config-ext-nacl)#deny ip 172.31.85.0 0.0.0.255 172.31.15.0 0.0.0.255
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#
Router(config-ext-nacl)#interface GigabitEthernet0/2.85
Router(config-subif)#ip access-group GUEST-FINANCE-BLOCK in
Router(config-subif)#
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#|
```

9.8 ACL Verification

After configuring the ACL, connectivity tests are performed to verify that the access-control requirement is working correctly.

Verification

The Guest-to-Finance ping should be **unsuccessful**, confirming that the ACL is blocking the restricted traffic.

A permitted connectivity test can also be performed from the Guest network toward an allowed destination to ensure that the ACL is not unnecessarily blocking all traffic.

```
Router#show access-lists
Extended IP access list GUEST-FINANCE-BLOCK
 10 deny ip 172.31.85.0 0.0.0.255 172.31.15.0 0.0.0.255
 20 permit ip any any

Router#
```

```
C:\>ping 172.31.15.10

Pinging 172.31.15.10 with 32 bytes of data:

Reply from 172.31.85.1: Destination host unreachable.
Reply from 172.31.85.1: Destination host unreachable.
Reply from 172.31.85.1: Destination host unreachable.
Reply from 172.31.85.1: Destination host unreachable.

Ping statistics for 172.31.15.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

9.9 ACL Result

The ACL implementation successfully provides access control between the Guest and Finance networks.

The configured extended ACL prevents traffic originating from the Guest VLAN (172.31.85.0/24) from reaching the Finance VLAN (172.31.15.0/24). At the same time, the permit rule allows other required network communication.

The verification results demonstrate that **Guest users cannot directly access Finance resources**, satisfying the network security requirement of the Westfold Textile Mills case study.

Conclusion

The Westfold Textile Mills network was successfully designed and configured to provide reliable communication between the Head Office, Weaving Plant, and Loomcraft Fabrics site. The implementation covered multi-area OSPF routing, EIGRP routing, OSPF-EIGRP route redistribution, EIGRP summarization, and dual-ISP connectivity using BGP.

Network security and address management were also implemented through Access Control Lists (ACLs), Network Address Translation (NAT), and Port Address Translation (PAT). The Finance and Guest networks were logically separated, with ACLs configured to restrict unauthorized access to Finance resources. NAT and PAT provided the required translation between the internal private network and external networks, while Static NAT enabled external access to the internal Web Server.

The configuration and verification performed throughout the project demonstrated the operation of routing, redundancy, address translation, and access control within the simulated enterprise environment. Overall, the case study provided practical implementation of networking and cybersecurity concepts using

Cisco Packet Tracer and demonstrated how these technologies can be combined to build a structured, secure, and interconnected enterprise network.

Project Outcome

The completed implementation meets the major networking and security requirements of the Westfold Textile Mills case study and provides a practical demonstration of enterprise network design, routing, access control, and address translation.